

ROLE OF INDUSTRY FACTORS IN FINANCING THE OUTWARD FOREIGN DIRECT INVESTMENT BY INDIAN MNES

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Abstract

The paper investigates the impact of industry environment of the home country (in which Indian parent firm operates) on financing their outward foreign direct investment (OFDI) for the period 2008-09 to 2013-14. Due to difficulty in empirically examining the flows within a multinational system there exists scant literature in this area. By employing random effects probit model we find that size and growth rate of the industry have important implications for OFDI financing by parent firm. Secondly, by including time effects, uniqueness of the industry to which the parent firm belongs significantly shapes the OFDI financing. Thirdly, parent firms are found to significantly rely on their own strengths than industry environment in financing the OFDI, lending support to the ownership advantage theory of international business. Finally, parent firm is found to follow industry norms in financing their OFDI. The study has implications for supply-side factors determining capital structure of firm and internal capital available to a multinational.

Keywords: outward foreign direct investment; financing; firm-specific factors; parent debt; industry-specific factors; random effects probit model; uniqueness

Introduction

Studies analyzing the capital structure of firm control for various industry factors. Also, the studies on capital structure of multinationals focus on the demand side factors that shape the capital structure of foreign affiliates of the parent firm by specifying the industry factors as control factors. We argue that the affiliate's capital structure is also affected by the industry to which the parent firm belongs i.e. supply-side factors. This has an impact on the financing decision of parent firm while financing its affiliates (via parent debt or parent equity). Although bankruptcy is a firm level phenomenon, Scott (1980) argues that industry structure is an important determinant of bankruptcy risk. Hence, industry factors have implication for provision of funds to foreign affiliates. By analyzing the industry factors that shape the capital

structure of foreign affiliates we move into the realm of outward foreign direct investment (OFDI).

Increasing outward foreign direct investment (OFDI) from emerging economies has garnered substantial attention. Many studies have been focussing on the determinants and motives of OFDI (Bruhn et al., 2016; Buckley et al., 2007; Deng, 2004; Elango & Pattnaik, 2007; Jayanthi, Sivakumar, & Haldar, 2016; Iammarino & Pitelis, 2000; Pradhan, 2004; Kumar, 2007; Witt and Lewin, 2007). Most of the studies (Gozzi, Levine, & Schmukler, 2010; Henderson, Jegadeesh, & Weisbach, 2006; Robbins & Stobaugh, 1972; Tripathi & Thukral, 2013; Tripathi & Thukral, 2017) that focus on the financing activities of multinational firms abroad are descriptive in nature. Tripathi and Thukral (2016) empirically examines the impact of host country determinants on OFDI financing and notes that the legal and financial environment in host country significantly shapes the OFDI financing decision by Indian MNEs. Tripathi & Thukral (2018) examines the impact of characteristics of emerging market's parent firm in shaping their OFDI financing decision. The present paper considers the industry environment of the parent firm in home country (industry factors) to analyze how it shapes their decision of financing the OFDI. Hymer (1976) recognized that FDI is a firm-level strategy rather than a capital-market financial decision. Therefore, we employ parent firm-level data to analyze the financing of outward foreign direct by considering industry environment followed by the parent-firm characteristics in shaping their firm's financing strategy in financing their OFDI.

With the increasing outward foreign direct investment by 'emerging dragons' (Mathews, 2010) we take up one such emerging economy – India that has been ranked 21st in OFDI in 2010 (UNCTAD, 2010). OFDI can be financed internally (internal capital market) by the MNE by providing debt (called parent debt) and equity (called parent equity), or externally, by raising finance outside the multinational system i.e. from external capital market (Tripathi & Thukral, 2016). We take up a case where Indian MNE (called parent firm) invests in their foreign affiliates, which is nothing but outward foreign direct investment from the perspective of parent firm. The investment may take the form of parent debt and parent equity, thereby tapping internal capital market. Internal capital market may have positive implications for business groups by overcoming financing constraints (Hoshi, Kashyap, & Scharfstein, 1991), called the bright side of internal capital market. However, various issues of internal capital market relating to tunnelling and propping activities of firms (Johnson, La Porta, López de Silanes, & Shleifer, 2000) results in inefficient utilization of internal capital market, called the dark side of internal capital market. From the perspective of foreign affiliates, the nature of funds provided by parent firm shapes their capital structure. The financing of OFDI (OFDI financing decision) of Indian parent firm may be shaped by industry factors.

It has been found that as industries globalize, firms in that industry are also pressured to globalize (Chang & Rosenzweig, 2001). Firms in oligopolistic market structures tend to react to the actions of the rival firms (Flowers, 1976; Knickerbocker, 1973; Vernon, 1966; Yu & Ito, 1988). We argue that the parent debt financing by a firm in the industry may be followed by the rival firms in that industry. With this in mind, the paper intends to determine the impact of parent's industry factors that shape OFDI financing decision of Indian parent firms.

Due to the nature of the dependent variable which is dichotomous and the inconsistent estimation of fixed effects in short panels, random effects probit model has been adopted in the study. The results indicate that size and industry growth rate of industry to which parent firm belongs significantly shape the financing of OFDI. However, with passage of time, it is the uniqueness of the industry that significantly shapes the financing decision of Indian parent firms. Thirdly, with inclusion of parent firm factors to the regression specification, we note that with time the importance of industry factors wither away and the firm rely on their own strengths in financing the OFDI. Lastly, results also indicate that Indian parent firms are found to follow industry norms in deciding the nature of finance to be provided to their foreign affiliates (OFDI).

Rest of the paper is organized as follows. Section two presents review of literature. Section three discusses industry-specific factors that have an impact on OFDI financing decision of Indian parent firms, followed by data and methodology in section four. Section five discusses the results. Finally section six concludes and highlights the managerial and theoretical implications of the study.

Review of Literature

Many studies relating to capital structure have been controlled for industry effects believing that industry effects could alter the capital structure (e.g., Doukas & Pantzalis, 2003). The type of industry and other characteristics of the industry have been shown to influence the extent of transferable profits (Geroski & Jacquemin, 1988) and hence the internal funds transferred. Geroski & Jacquemin (1988) finds that industry's growth rate is positively related to the extent of firm profitability and consequently to the internal financial flows. Desai, Foley, and Hines Jr. (2004) and Aggarwal and Kyaw, (2008) also control for industry effects while studying host country factors in determining borrowing from parent. Shin and Park (1999) finds Korean chaebols invest more than the industry average, regardless of their investment opportunities compared to non-chaebol firms in the same industry. Aulakh and Mudambi (2005) finds that capital investment of unrelated segments of conglomerate firms is less responsive to the industry's Tobin's Q than of standalone firms. They find that flows from subsidiary to the headquarter decreases with growing concentration and growth in the industry to which the subsidiary belongs, though the relationship is not significant. Aggarwal and Kyaw (2008) highlights that the country level factors as determinants of affiliates' capital structure are more important than the industry level factors. They examine determinants of affiliates' overall leverage (total debt ratio) of foreign affiliates operating in 62 countries for sample years- 1989, 1994 and 1999. They run industry level regression to further examine whether multinational affiliates conform to local capital structure norms. When industry level regressions are estimated, political risk, private credit and bank variables are no longer significant. By controlling for national institutional characteristics such as determinants of capital structure, they find that influence of local capital structure norms disappears. Desai et al. (2004) analyzes majorly the country level determinants of capital structure of foreign affiliates of US multinational firms, controlling for industry effects. Buettner, Overesch, Schreiber, and Wamser (2009) finds that local tax burden exerts important effects on the leverage (external and internal debt) of foreign affiliate of a German multinational. The large size or cash flow (turnover)

improves the access to external capital and reduces access to internal capital. The results are robust if industry dummies are included to control for heterogeneity among affiliates. De Jong, Kabir, and Nguyen (2008) also finds that determinants of capital structure across countries remain materially robust to the inclusion of industry dummies. Akbel and Schnitzer (2011) examines industry-level patterns and observes that MNEs operating in industries with long-term payoff periods prefer a centralized (local) borrowing structure, whereas MNCs in industries with short-term pay-off periods prefer a decentralized borrowing structure (parent borrowing).

Above studies that analyze the impact of industry factors, consider the industry to which the foreign affiliate belongs. However, the characteristics of industry in which parent firm operates also impact the capital structure of the foreign affiliate and eventually the financing decision of the parent firm. External lenders desirous of insulating themselves against impending bankruptcy of foreign affiliate, makes them to inquire about the parent firm also. Turmoil in the parent's industry may jeopardize the operation of parent. Affiliates of such parent are viewed with scepticism by external lenders that refrains them to lend unless at a higher cost. Industry structure is an important determinant of bankruptcy risk (Scott, 1980). For example, reduced sales growth in an industry may reduce demand of products in that industry forcing small firms to exit from the industry. This is of great concern to the external lenders while lending to affiliates of parent firm belonging to such industry.

Objectives

In light of scant literature that analyzes industry factors that impact parent financing (or, parent debt), the present paper answers the research question - what industry-specific advantages (characteristics of the industry to which the parent firm belongs) propels Indian parent firms to provide parent debt financing to its foreign affiliates. Parent debt financing provided is nothing but outward foreign direct investment by Indian parent firms. By performing industry-focused work, the paper contributes to the literature of internal capital market that attempts to capture industry effects. The study also has implications for supply-side determinants of capital structure of foreign affiliates.

Theoretical Framework: Industry-Specific Factors Affecting OFDI

The benefit of analyzing rich industry-specific data sets and interpreting results in specific industry context have been highlighted by various studies (Berger, Miller, Petersen, Rajan & Stein, 2000; Khanna & Tice, 2001; Mullainathan & Scharfstein, 2001). The present section discusses factors constituting the industry environment in which parent firm operates and their probable impact on the OFDI financing decision.

A high sales growth of industry to which parent firm belongs can take up risk while investing in its affiliates via parent equity. A high growth rate may signal external creditors about the sustainability of its foreign affiliate that increases their propensity to lend to the affiliates at a low cost. Availability of funds from external market reduces the provision of parent debt to the foreign affiliate. Thus, we conjecture a negative impact of industry's growth rate on the

propensity of providing parent debt to foreign affiliates. The present study measures home country's industry growth rate (ISalesgr) by sales growth over past 3 years.

The larger the size of the firm, the lesser is the severity of asymmetric information for outside investors (Myers & Majluf, 1984). Large firms are less risky (Harris & Raviv, 1991; Kwok & Reeb, 2000; Olibe, Michello, & Thorne, 2008) and hence it is easier to obtain debt by a large firm. Debt ratios of larger firms are less affected by financial distress (Smith & Watts, 1992). These firms are less risky and can assume further risk by providing parent equity. Also, an industry which is large is perceived by external lenders to be more stable and resilient to financial distress. This causes them to grant debt to affiliate whose parent firm belongs to a large industry. This may reduce reliance of foreign affiliate on parent debt, indicating a negative impact of size of the industry (to which parent firm belongs) on the propensity of providing parent debt by parent firm to their foreign affiliates. The current study uses log of sales as a proxy for industry's size, operationalized by the variable, ISize.

We postulate that industry's capital structure norms have an impact on OFDI financing. There are studies that suggest that capital structure of foreign affiliates is influenced by local (host country) capital structure norm (Sekely & Collins, 1988; Shapiro, 1978; Stonehill & Stitzel, 1969). However, Naumann-Etienne (1974) argues that target debt ratios of US MNEs depend on risk perceptions of investors in US suggesting that capital structure of foreign affiliates may be in conformity with parent firm. Financing decision of parent indirectly decides whether the capital structure of foreign affiliate conform to local norms of the host country, or conform to parent company norms, or a mix of these two to take mileage of opportunities offered by host country environment (e.g. taxes, credit availability, conducive political environment etc.) to minimise the overall cost of capital.

The financing decision of parent depends on whether and to what extent the parent firm is responsible for meeting the financial obligations of its foreign affiliates. When the onus of affiliate's obligations lies on parent, the affiliate's debt-equity norm does not vary from that of its parent unless the parent is willing to allow its affiliate to default. At the other end of the continuum, when the parent does not take the responsibility of meeting obligations of its affiliate and would allow it to default, the affiliates debt-equity norm is influenced by its own financing decisions in host country guided by reducing its default risk. However, a parent allowing the default of its affiliate is just illusory in practical world as the parent cannot elude the impact of such dire consequences. Blindly following local (host) country financing norm may be too naïve for the MNE as a whole when the local country financial markets are not well-developed and stable. So, the parent's debt-equity structure is also relevant for determining the capital structure of foreign affiliates. Thus, the financing decision of parent to finance its affiliates is shaped by the parent's capital structure. The parent, in turn, is found to comply with capital structure norm of the industry to which it belongs. Thus, industry to which the parent belongs has a role in explaining financing decision of parent.

Stonehill and Stitzel (1969), and Shapiro (1978) argue that affiliate's capital structure reflects local country financing norms. Eiteman, Stonehill, and Moffett (2010) notes following advantages that accrues to the foreign affiliate by conforming to local financial structure norms.

Firstly, it reduces skepticism of lenders in host country. If the debt ratio is higher than host country's capital structure norms, the foreign affiliate does not have enough economic capital to avoid default and if the debt ratio is too low, the foreign subsidiary tries to escape the effect of the local monetary policy. Secondly, it facilitates comparison of return of equity of affiliates to the local firms. Thirdly, it reminds management not to misallocate resources because they become more cautious about the efficiency of allocating resources in case the cost of capital is high in host country.

Eiteman et al. (2010) provides following reasons for an affiliate resorting to parent's local norms. Since, affiliates are expected to have a competitive advantage over local firms in overcoming imperfections in foreign capital markets so it may be unreasonable to conform to local norms to adopt a suboptimal capital structure. Also, in certain cases, conformance to local capital structure norms increases the cost of capital and financial risk for the MNE as a whole. Most importantly, the external lenders not only take into account the affiliate before lending but also keep a check on the parent and the other affiliates worldwide. In this sense, local debt ratios are meaningless for them. However, Naumann-Etienne (1974) argues that affiliate resorts to parent's capital structure in order to meet the expectations of investors in the parent country. They found this in case of US MNEs.

Aggarwal and Kyaw (2008) study the impact of local (host country) capital structure norms in affecting the leverage of firms in a particular industry across countries where affiliates are situated. They find that the affiliate's leverage ratios are similar to local norms only for some industries. They further show that influence of local capital structure norms on industry leverage disappear when controlled for host country variables. They performed such industry analysis for the year 1999.

However, the present paper attempts to test if capital structure norm of parent's industry in the home country impacts its decision of providing parent debt to its affiliate abroad. An affiliate not following the parent's industry norm may be viewed with scepticism by the lenders in the host country and may increase the cost of providing external funds. To minimise these potential conflicts the affiliate may want to maintain their capital structure in alignment with that observed by other firms in the industry of home country. A positive relation between the capital structure norm followed in home country's industry and the parent debt financing would reduce such skepticism that may facilitate affiliate to raise loan from host country. A negative relation would accentuate incredulity in external investors and hence may reduce debt-raising in host country that may steer the foreign affiliate to tap internal capital market. So, if onus of affiliates' obligations lies on parent, then affiliates' debt-equity norm does not vary much from parent's capital structure norm. The present study envisages a positive relationship between industry's capital structure norm and parent debt-financing. Industry's capital structure norm (ICapStr) is measured by Long-term debt/total capital.

Industry that spends more on advertising is believed to be investing in reputational or brand assets. Firms of such industry possess unique information that gives them competitive advantage over firms of other industry. The industry would be reluctant to divulge out such information to external lenders and increase their risk. For achieving this, they intend to have a

close control on operations of affiliate by purchasing their equity, indicating that an industry which is high on advertising intensity tends to provide less parent debt to their affiliates abroad. The present study measures advertising intensity of the industry (IAdvInt) by expenditure on advertisement by the industry scaled by total sales of the industry.

An industry that invests more in research and development expenditures is perceived to be riskier. Such R & D intensive industries carry less debt in their capital structure. Thus, external lenders impound a higher risk premium. This reduces the reliance of foreign affiliate on external lenders to raise loans. Coupled with moral hazard problems of investing in a foreign affiliate by external lenders and with a view to reduce the risk-avoiding behaviour of managers at the affiliates to prune R & D investments, parent firms tend to provide parent debt to the foreign affiliates. R&D intensity of the industry (IRndInt) is measured by expenditure on R&D expenditure by the industry scaled by total sales of the industry.

An industry that expands internationally tends to have a fair understanding of the external capital market. Also, the external lenders are not reluctant to lend to firms that have some experience and transactions with the external market. We, therefore, conjecture that industry which has a greater international experience tends to provide less parent debt to foreign affiliates. The study measures international experience of the industry (IExpEtoSales) by export earnings of the industry scaled by its total sales.

Data and Methodology

For assessing the impact of parent's industry factors on OFDI financing, the sample includes 88 parent firms that have undertaken outward foreign direct investment in all six years of the sample period, 2008-09 to 2013-14. The dataset is provided by the Reserve Bank of India (RBI) that reports the outward direct investment by Indian firms. It reports the outflows from India in the form of debt and equity. The sample includes those firms that have invested incessantly in all six years. These firms belong to 40 industries as per National Industrial Classification (NIC). These were then clubbed into six broad industries at three-digit NIC level. These are viz., Industrials, Fast Moving Consumer Durables (FMCG), Health care, Information and Communication Technology (ICT), Energy and Utilities, and Materials.

For data on these industries we referred BSE Industrial indices from Prowess maintained by Centre for Monitoring Indian Economy (CMIE) that reasonably represent a particular industry. Data related to the explanatory variables of all firms in that index was taken and then median of the respective explanatory variables for each industry was taken. The operational definitions for explanatory variables and the BSE Industrial indices used to measure explanatory variables are shown in Table 1.

Dependent Variable

Following Aggarwal and Kyaw (2008), relative proportion of debt in parent financing i.e. ratio of parent debt provided to the affiliate to total parent investment (parent debt plus parent's equity investment) in the affiliate is used as the dependent variable. Since most of the

firms were either debt oriented or equity oriented (i.e. the ratio of parent debt to total financing was mostly 0 or 1), we converted these ratios into dichotomous variable with one indicating firms whose parent debt ratio to total parent financing is greater than or equal to the median of parent debt to total financing ratio, and zero otherwise.

Independent Variables

Following Bernard and Jensen (2004), Bhaumik, Driffield, and Pal (2010), Roberts and Tybout (1997) a panel structure with lags is constructed to attest endogeneity problems of the explanatory variables. Lagging the independent variables by one time period captures the fact that firms rely on past data to make their investment decisions in financing their affiliates.

Methodology

Since the estimation of fixed effects in short panels is inefficient, and the loss of observations in the dependent variable when $\sum_{i=1}^T y_{it} = 0$ or 1 (i.e. no variation in y_{it} over t), random-effects probit model is adopted (Cameron & Trivedi, 2005; Greene, 2012).

The two models adopted in the study are:

Model 1: OFDIF = f(Parent's Industry Characteristics)

Model 2: OFDIF = f(Parent's Industry Characteristics, Parent's Firm Characteristics)

For the above models, the specification for random effects probit model can be stated as:

$$\Pr(\text{OFDIF}_{it}) = 1 | x_{it}, \alpha_i, \beta x_{it} = \alpha_i + \beta x_{it} + \varepsilon_{it}; i = 1 \text{ to } 88; t = 1 \text{ to } 6$$

Besides capturing the parent firm-specific heterogeneity we also control for omitted variables that are constant over firms but vary across time (Altomonte, 2000; Altomonte & Guagliano, 2003; Amighini, Rabellotti, & Sanfilippo, 2011; Martin & Salomon, 2003; Greenaway, Guariglia, & Kneller, 2007). We control for these time effects by introducing five dummy variables for six year time period to avoid 'dummy variable trap'.

In case of a probit model, the coefficients cannot be interpreted in a direct way. One needs to interpret the marginal effects of the regressors, that is how the probability of the outcome variable change when the value of a regressor changes, *ceteris paribus*. Correspondingly, the coefficients reported in the tables 2, 3 and 4 are the Marginal Effects that have been explained in the subsequent section. For probit model, Marginal Effects is given by:

$$ME_j = \frac{\partial E[y]}{\partial x} = \phi(x'\beta)\beta_j$$

Results and Analysis

Table 2 shows the results of the impact of the change in various industry factors on the probability of parent debt financing. Column 1 indicates that the probability of providing parent debt by parent firm is significantly increasing in industry growth rate (significant at 5%), and industry size (significant at 1%). Empirical results indicate that for every increase in growth in sales of the industry, the propensity to provide parent debt to affiliates vis-à-vis parent equity

increases. Such firms could have taken risk because the industries that are big and experience growth in sales might have accumulated sufficient internal funds over a period of time. On the contrary, these firms are seen investing via non-risky capital in foreign affiliates. This point towards the risk averse nature of firms in India with regard to investing abroad in their affiliates. This is a hint to inefficient use of internal capital market of the multinational system. The highly significant p-value for Wald chi-square statistic in Table 2 indicates that all the coefficients jointly are significantly different from zero and hence the overall fit of the model is robust. Additional regressions are also reported (columns 2 through 7) to check the consistency of the results in column 1 in terms of significance and sign of the variables, specifically due to the small time period of the study. The results in these columns reinforce the findings of Column 1, the main model, that industry growth rate and industry size are the important drivers that affect parent debt financing decision.

All the columns (1 through 7) consistently show that the factors – industry's capital structure norm, advertising intensity, R & D intensity, and international experience do not emerge as significant factors but provides cues regarding their probable impact on parent debt financing. Increase in likelihood of providing parent equity in response to high advertising intensity of the parent's industry indicates the intention of parent firm to closely control the operations of its affiliate to prevent it from divulging out unique information to external lenders. An industry wary of defaulting on loans that may tarnish its image increases the confidence of external lenders thereby increasing their propensity to lend. This reduces the propensity to provide parent debt. The study also notes that parent's industry that carries high debt in its capital structure resorts to more parent debt financing vis-à-vis parent equity financing. Increasing debt in firms of an industry faces high liquidity risk and hence is reluctant to raise further risk by providing risky capital – parent equity to its foreign affiliate. Also, from external lender's point of view, an industry experiencing financial risk and thereby having high chance of bankruptcy is viewed with scepticism by external lenders. This induces the parent firm's industry to invest via more parent debt. Results also indicate that propensity of providing parent debt decreases with increase in international experience of the parent's industry. This may be because parent firm that has some experience with foreign market tend to rely on external capital market.

Impact of Time and Industry Effects

Incorporating time effects in the empirical results obtained above, we note that the importance of industry growth rate and size of the industry to which parent firm belongs in shaping financing of OFDI gets subsumed (see Table 3 column 1). The overall fit of the model is robust as indicated by significant value of the Wald chi-square statistic. The R & D intensity of the industry emerges as a significant factor (significant at 10%) affecting the financing decision of Indian parent firms. This indicates that with the increasing expenditures on R & D in an industry reduces the likelihood of providing parent debt to their affiliate abroad. The reason can be attributed to the need of controlling the operations of the affiliate that would deter them to divulge out the information about the uniqueness of the firm to the external creditors. Thus, controlling for time effects, we note that Indian firms do not venture abroad mainly on the basis of how the industry is growing, or the size of the industry. Rather, it is the uniqueness of the industry that significantly shapes the financing decision of the Indian parent firm to finance its affiliates. Results demonstrate that

propensity of firms to provide parent debt to foreign affiliates reduces. Earlier studies found that at industry level, market size and growth rates were important determinants of firm's survival (Mata and Portugal, 1994). However, recent study by Cefis and Marsili (2006) finds a positive impact of innovation on firm's survival probability with small and young firms benefitting the most. They find this effect to be pronounced over longer time periods. Prominence of innovation in firm's survival has also been found in case of Chinese firms (Zhang & Mohnen, 2013). Thus, over a period of time it is the uniqueness of industry than industry's size and its growth rate that significantly guides Indian MNEs to take their financing decision by reducing the likelihood of providing parent debt. Regressions in columns 2 through 7 are presented to show the consistency of results in terms of sign and significance of coefficients reported in column 1, the main model of Table 3.

Impact of Industry and Parent Firm Level Factors

Impact of industry and parent firm characteristics on parent debt financing have been examined in Table 4. When the impact of industry factors is studied along with firm factors, significance of industry factors gets subsumed as was observed in Table 2 and 3. However, parent characteristics – age, capital structure, R & D intensity, growth opportunities available to the firm along with systematic risk and operating risk emerge as significant determinants that affect the propensity of providing parent debt to foreign affiliates (refer Table 4). None of the parent's industry determinants emerges significant, indicating that when time effects are present, industry characteristics do not significantly influence the decision of parent firm to finance their OFDI. This indicates that Indian parent firms rely more on their own strengths than industry factors in financing their outward foreign direct investment (see Table 4).

It is also noted that all industry-related factors – capital structure norm, size, growth rate, R & D intensity, advertising intensity, internationalization of industry bear same sign as obtained when OFDI financing is regressed only on parent-firm factors¹. This indicates that parent firms are seen to follow the same norms as followed in their respective industry. The firms in an industry imitate other firms to constrain competition (Porter, 1979). The finding also supports various studies suggesting that firms imitate the moves of other rival firms (Chen & MacMillan, 1992; Karnani & Wernerfelt, 1985; Knickerbocker, 1973; Yu & Ito, 1988). Time dummies that emerge significant in the estimations are a testimony to the importance of including time effects in the analysis.

Conclusion

Due to difficulty in empirically examining the capital flows within a multinational system there is scant literature that analyzes the factors determining financing of OFDI from a country. With increasing OFDI from developing economies and their changing character, we take up a case of one such economy – India. Indian parent firms can finance their foreign affiliates by internal funds in the form of parent debt or parent equity. By analyzing OFDI by Indian MNEs the study also throws light on internal capital market of multinational system on

¹ Results available on request from authors

one hand and supply side factors of the capital structure of foreign affiliates, on the other hand.

The present paper explains the role of industry environment of the home country that shapes the likelihood of Indian parent firms to provide parent debt financing to their affiliates abroad. The study notes that size and growth rate of industry (to which parent firm belongs) significantly influence its OFDI financing decision. Owing to the erratic nature of FDI, the study also controls for time effects and finds that it is the size and growth rate of industry subsumes and it is the uniqueness of the industry to which firm belongs that significantly affects OFDI financing of Indian MNEs by increasing likelihood of providing parent debt to foreign affiliates. It is important for firms to innovate in order to survive in the long term which is evident in the present study that highlights the role of innovation within the industry in guiding firm's OFDI financing decision.

By including firm-specific factors (parent firm's characteristics) along with industry-specific factors, the study notes that significance of industry environment subsumes. The parent firms are found to rely more on firm-specific advantages (i.e. ownership advantages) than factors of the industry to which it belongs. Firm factors viz., age, capital structure of the firm, expenditure on R & D, growth opportunities available along with systematic risk and operating risk significantly shape their financing strategies. The signs of these factors in the analysis are same as those that appear when industry factors are also included, highlighting the fact that parent firms are found to follow the industry norms when financing their outward foreign direct investment, lending support to imitation theories of firms. Thus, Indian MNEs may take cues from the behaviour of rival firms in the industry but must rely on their firm-specific advantages while financing their OFDI. From the perspective of foreign affiliates, these are factors that also shape the capital structure of foreign affiliates.

Managerial Implications

The general theory of multinational enterprise mainly attempt to explain what motivates the firms to go abroad, what enables them to do so, why they undertake different forms of investment abroad. The theories do not account for the finance-specific factors. The inclusion of these finance-specific factors in the investment decision has also been emphasized by Oxelheim, Randoy, and Stonehill (2001). It is important to study finance-specific factors because one of the reasons that deter firms in undertaking FDI is the paucity of funds to finance FDI (Welch & Benito, 2007). The present study explains the behaviour of Indian MNEs in taking the financing decision of their OFDI and it becomes worthwhile to weave such financial framework in the internationalization theories.

Besides the theoretical underpinnings, the study also has implications for managers at parent firm. Industry innovativeness must be taken as a guide by the Indian MNEs to finance their OFDI. In the present world driven by innovation, even smaller firms that are innovative are the

ones that thrive and compete for long. As India is increasingly becoming a top global innovator² that has been found to be investing more in innovation even in times of recession (Radjou, 2008), firms must be able to leverage its innovation even in shaping their OFDI financing decision.

Notwithstanding that parent equity is characteristically a risky capital, is closely controlled by host government and is heavily taxed in host, parent firms are more likely to provide equity. This is because parent equity provides parent with control and monitoring rights over the affiliates that could prevent the latter to divulge out the information to external lenders and other stakeholders in the host country. The covenants in the external loans raised by affiliates may leak out the information to external lenders about the parent firms that may thwart its uniqueness and eventually erode its competitive advantage. This will help managers to adopt a sustainable long-term strategy and efficiently utilize its internal capital.

Further, the study suggests that Indian MNEs must rely on their own strengths while financing their OFDI than just following the strategies by other firms in the industry. More than imitating their rivals (as suggested by literature), a sustainable strategy of a firm must emanate from the strong firm-specific advantages. Following their strengths and not blindly following their peers shall shape an effective financing strategy of the Indian MNEs.

Thus, the present study provides cues to the significant determinants of financing the OFDI from emerging economies, a phenomenon that has been catching attention recently in literature. The main limitation of the paper is the smaller number of years in the analysis due to recent availability of data on the components of OFDI categorized into debt and equity by Reserve Bank of India. Inclusion of more number of years will heighten the richness of study. Studies on financing pattern on outward FDI from India is in their infancy and extending the present study to financing strategies of firms from other emerging economies and consequently generalizing the findings is an interesting avenue for future research. This may pave way for a theory of multinational enterprise that incorporates the financing dimension of internationalization of firms. The study, thus, attempts to set a stage to explore more issues in the area of financing the outward foreign direct investment from India.

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² Five Indian companies have been featured in 2016 Forbes's list of 'The World Most Innovative Companies'. India is at 66th position in Global Innovation Index 2016 ranking and tops in Central and South Asia.

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Table 1: Operational definitions of explanatory variables

Variable name	Proxy	Definition
Industry Size	ISize	Log of sales
Capital structure	ICapStr	Long-term debt/total capital
R&D intensity	IRndInt	R&D expenditure/sales
Advertising intensity	IAdvInt	Advertisement expenditure/sales
International experience	IExpEtoS ales	Export earnings/sales
Growth rate	ISalesgr	Sales growth over past 3 years
FMCG industry	Dfmcg	Dummy (BSE Industrial index - Consumer Discretionary and FMCG)
Energy & Utilities	Denergy	Dummy (BSE Industrial index- Energy, and Utilities)
Healthcare industry	Dhealth	Dummy (BSE Industrial index - Health care)
Information, Communication, and Technology	Dict	Dummy (BSE Industrial index - Information Technology, Telecom, and Media)
Materials	Dmat	Dummy (BSE Industrial index - Materials)
Industrials	Dind	Dummy (BSE Industrial index - Industrials)

Source: Prepared by the authors

Table 2: Robust Random Effects Probit Model Results: Impact of Industry factors on parent's financing decision (without time and industry effects)

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ICapStr	4.514 (0.242)	4.714 (0.219)	4.275 (0.252)	4.476 (0.228)	4.394 (0.252)	4.095 (0.267)	4.570 (0.232)
IRndInt	-35.504 (0.541)		-31.789 (0.591)		-37.655 (0.491)	-34.355 (0.542)	
IAdvInt	-66.357 (0.531)	-59.676 (0.580)			-70.331 (0.498)		-64.363 (0.544)
IExpEtoSal es	-0.722 (0.808)	-0.949 (0.735)	-0.968 (0.740)	-1.148 (0.677)			
ISalesgr	0.005** (0.038)	0.005** (0.037)	0.005** (0.038)	0.005** (0.037)	0.005** (0.038)	0.005** (0.038)	0.005** (0.037)
ISize	5.649*** (0.000)	5.538*** (0.000)	5.500*** (0.000)	5.414*** (0.000)	5.597*** (0.000)	5.418*** (0.000)	5.459*** (0.000)
Dfmcg	1.690 (0.193)	1.572 (0.232)	0.920* (0.069)	0.883* (0.072)	1.761 (0.160)	0.953* (0.054)	1.656 (0.197)
Dhealth	1.910 (0.146)	1.185 (0.280)	1.771 (0.197)	1.127 (0.307)	1.708 (0.194)	1.482 (0.285)	0.855* (0.084)
Dict	2.958* (0.093)	2.631 (0.160)	2.788 (0.117)	2.506 (0.178)	2.572** (0.015)	2.246** (0.031)	2.087** (0.018)
Denergy	-1.240* (0.093)	-1.175* (0.160)	-1.161* (0.117)	-1.109* (0.178)	-1.193* (0.015)	-1.090* (0.031)	-1.107* (0.018)

	(0.070)	(0.087)	(0.085)	(0.098)	(0.076)	(0.096)	(0.093)
Dmat	-1.039	-1.042	-1.069	-1.069	-1.021	-1.047	-1.018
	(0.117)	(0.116)	(0.110)	(0.110)	(0.121)	(0.116)	(0.122)
	-	-	-	-	-	-	-
Constant	23.136**	22.802**	22.601**	22.351**	22.963**	22.327**	22.546**
	*	*	*	*	*	*	*
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Obs.	528	528	528	528	528	528	528
No. of firms	88	88	88	88	88	88	88
Log Lik	-287.6	-287.9	-287.9	-288.1	-287.7	-288	-287.9
Wald chi2	41.95	41.65	41.61	41.35	41.97	41.59	41.63
Pval	1.65e-05	8.66e-06	8.79e-06	4.32e-06	7.60e-06	3.91e-06	3.83e-06

Robust pval in parentheses; *** p<0.01, ** p<0.05, * p<0.1

Source: Authors' calculations

Table 3: Robust Random Effects Probit Model Results: Impact of industry factors on OFDI financing (with time and industry effects)

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ICapStr	4.823	6.084	4.764	6.026	4.994	4.980	6.126
	(0.324)	(0.213)	(0.322)	(0.210)	(0.312)	(0.306)	(0.213)
	-		-		-	-	
IRndInt	111.475*		111.478*		105.558*	105.590*	
	(0.090)		(0.090)		(0.078)	(0.076)	
IAdvInt	-8.993	-8.231			-1.902		-6.329
	(0.941)	(0.945)			(0.987)		(0.957)
IExpEtoSales	1.353	0.376	1.322	0.349			
	(0.681)	(0.900)	(0.684)	(0.905)			
ISalesgr	0.003	0.003	0.003	0.003	0.003	0.003	0.003
	(0.315)	(0.199)	(0.329)	(0.213)	(0.288)	(0.302)	(0.195)
ISize	0.479	1.336	0.449	1.310	0.625	0.618	1.364
	(0.845)	(0.576)	(0.854)	(0.585)	(0.798)	(0.800)	(0.569)
Dfmcg	0.347	0.355	0.242	0.259	0.226	0.204	0.321
	(0.820)	(0.812)	(0.710)	(0.689)	(0.880)	(0.753)	(0.825)
Dhealth	1.348	-0.509	1.337	-0.519	1.703	1.699	-0.379
	(0.346)	(0.693)	(0.360)	(0.690)	(0.215)	(0.231)	(0.642)
Dict	0.774	0.429	0.749	0.405	1.486	1.477	0.641
	(0.687)	(0.827)	(0.699)	(0.837)	(0.212)	(0.209)	(0.579)
Denergy	-0.246	-0.351	-0.232	-0.338	-0.337	-0.334	-0.376
	(0.759)	(0.657)	(0.771)	(0.666)	(0.673)	(0.672)	(0.633)
Dmat	-1.061	-1.117	-1.064	-1.119	-1.094	-1.094	-1.127
	(0.156)	(0.130)	(0.154)	(0.130)	(0.145)	(0.144)	(0.128)

Dt1	-1.894** (0.021)	1.528** (0.048)	-1.899** (0.020)	1.532** (0.048)	-1.878** (0.023)	-1.879** (0.022)	1.529** (0.048)
Dt2	-1.134* (0.080)	-0.911 (0.146)	-1.139* (0.078)	-0.915 (0.146)	-1.128* (0.082)	-1.129* (0.080)	-0.914 (0.145)
Dt3	-0.608 (0.288)	-0.512 (0.364)	-0.608 (0.288)	-0.512 (0.364)	-0.629 (0.275)	-0.629 (0.275)	-0.520 (0.356)
Dt4	-0.497 (0.161)	-0.324 (0.342)	-0.503 (0.160)	-0.330 (0.347)	-0.514 (0.148)	-0.515 (0.151)	-0.332 (0.321)
Dt5	-0.443** (0.042)	-0.386* (0.072)	-0.448** (0.043)	-0.390* (0.078)	-0.464** (0.033)	-0.465** (0.036)	-0.393* (0.062)
Constant	-2.026 (0.842)	-6.005 (0.545)	-1.907 (0.851)	-5.899 (0.554)	-2.528 (0.805)	-2.500 (0.806)	-6.085 (0.541)
Obs.	528	528	528	528	528	528	528
No. of firms	88	88	88	88	88	88	88
Log Lik	-279.8	-281.7	-279.8	-281.7	-279.9	-279.9	-281.7
Wald chi2	52.62	49.77	52.61	49.76	52.28	52.28	49.73
Pval	8.70e-06	1.31e-05	4.47e-06	6.69e-06	5.09e-06	2.52e-06	6.78e-06

Robust pval in parentheses; *** p<0.01, ** p<0.05, * p<0.1

Source: Authors' calculations

Table 4: Robust Random Effects Probit Model Results: Impact of parent firm-level factors and industry factors on OFDI financing (with time and industry effects)

Variables	(1)	Variables	(1)
ProfitROA	-0.388 (0.791)	Dhealth	2.477 (0.108)
Size	0.082 (0.701)	Dict	1.824 (0.356)
TobinsQ	0.154** (0.046)	Denergy	-0.191 (0.828)
CapStr	1.755** (0.021)	Dmat	-1.719** (0.032)
DRndInt	-0.490* (0.075)	Dt1	-2.264** (0.014)
DAdvInt	-0.495 (0.102)	Dt2	-1.268* (0.072)
DExpESales	0.015 (0.951)	Dt3	-0.749 (0.236)
Age	-0.013* (0.098)	Dt4	-0.608 (0.153)
Beta	0.983** (0.012)	Dt5	-0.512** (0.041)
DOL	-0.002*	Constant	-1.447

Variables	(1)	Variables	(1)
	(0.098)		(0.902)
DForHold	0.228		
	(0.410)	Obs.	497
ICapStr	3.324	No. of	
	(0.501)	firms	83
IRndInt	-107.244	Log Lik	-243.7
	(0.136)	Wald chi2	64.39
IAdvInt	-25.195	Pval	6.82e-05
	(0.856)		
IExpEtoSales	-0.668		
	(0.840)		
ISalesgr	0.003		
	(0.230)		
ISize	0.159		
	(0.954)		
Dfmcg	0.892		
	(0.609)		

Robust pval in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Source: Authors' calculations